

SOLIDWORKS API (.NET VBA C#) — Reference

(37/337): ICamera-ICavityFeatureData

SOLIDWORKS API Documentation (converted from CHM)

ICamera (Type)

ICamera Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : ICamera Interface

Allows access to the camera feature.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface ICamera
```

Visual Basic (Usage)



```
Dim instance As ICamera
```

C#

```
public interface ICamera
```

C++/CLI

```
public interface class ICamera
```

Visual Basic for Applications (VBA) Syntax

See [Camera](#).

Example

[Insert and Rotate Camera \(C#\)](#)
[Insert and Rotate Camera \(VB.NET\)](#)
[Insert and Rotate Camera \(VBA\)](#)

Accessors

[IFeature::GetSpecificFeature2 Method](#)
[IModelDocExtension::GetCameraById Method](#)
[IModelDocExtension::GetCameraDefinition Method](#)
[IModelDocExtension::InsertCamera Method](#)
[IModelView::Camera Property](#)

Access Diagram

[Camera](#)

See Also

[ICamera Members](#)
[SolidWorks.Interop.sldworks Namespace](#)

ICamera Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : ICamera Interface

For a list of all members of this type, see [ICamera members](#).

Public Methods

Name	Description
 GetFocalDistance	Gets the camera's focal distance.
 GetPointOfInterestDistance	Gets the distance of the point of interest from the camera.
 GetPosition	Gets the current position of the camera.
 GetPositionCartesian	Gets the Cartesian coordinates for the camera position.
 GetPositionEntity	Gets the entity to which the camera is attached.
 GetPositionSpherical	Gets the spherical coordinates for the camera position relative to the target, based on a sphere around the Z axis with the zero (0) point at the target.
 GetTargetPointEntity	Gets the target point on the entity for the camera.
 GetUpVector	Gets the camera's up direction.
 GetViewVector	Gets the direction in which the camera is looking.
 SetPositionCartesian	Sets the Cartesian coordinates for the camera position.
 SetPositionEntity	Sets the entity to which the camera is attached.
 SetPositionSpherical	Sets the spherical coordinates for the camera position relative to the target, based on a sphere around the Z axis with the zero (0) point at the target.
 SetTargetPointEntity	Gets the target point on the entity for the camera.

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See Also

[ICamera Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

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ICamera Interface Members

[See Also](#) [Properties](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : ICamera Interface

The following tables list the members exposed by [ICamera](#).

Public Properties

Name	Description
 AspectRatio	Gets the aspect ratio (width/height) of the field of view rectangle for the camera. NOTE: This property is a get-only property. Set is not implemented.
 DepthOfFieldEnabled	Gets whether depth of field effects are enabled or disabled. NOTE: This property is a get-only property. Set is not implemented.
 DepthOfFieldOffset	Gets the approximate distance from the camera's focal plane to point where focus is lost. NOTE: This property is a get-only property. Set is not implemented.
 FieldOfViewAngle	Gets the camera's horizontal angle of the field of view. NOTE: This property is a get-only property. Set is not implemented.
 FieldOfViewDepth	Gets the camera's depth of the field of view. NOTE: This property is a get-only property. Set is not implemented.
 FieldOfViewHeight	Gets the camera's height of the field of view. NOTE: This property is a get-only property. Set is not implemented.
 ID	Gets the ID for this camera.
 LockCameraPosition	Gets or sets whether the camera position is locked.

 Perspective	Gets whether the camera is in perspective mode or not. NOTE: This property is a get-only property. Set is not implemented.
 Pitch	Gets or sets the pitch (up or down angle) of a floating camera .
 PositionType	Gets or sets the camera position type.
 RotationRollAngle	Gets or sets the camera's roll angle.
 RotationRollBySelection	Gets or sets whether you can select the camera's rotation roll direction.
 RotationRollEntity	Gets or sets the entity (line, edge, face, or plane) that defines the camera's rotation up direction.
 RotationRollFlipDirection	Gets or sets whether to flip the direction of the camera 180.
 TargetPointBySelection	Gets or sets whether you can set the target point.
 TargetPointPosition	Gets or sets the position of the target point.
 Type	Gets or sets the type of camera.
 Yaw	Gets or sets the yaw (side-to-side angle) of a floating camera .

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Public Methods

Name	Description
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 GetPosition	Gets the current position of the camera.
 GetPositionCartesian	Gets the Cartesian coordinates for the camera position.
 GetPositionEntity	Gets the entity to which the camera is attached.
 GetPositionSpherical	Gets the spherical coordinates for the camera position relative to the target, based on a sphere around the Z axis with the zero (0) point at the target.
 GetTargetPointEntity	Gets the target point on the entity for the camera.
 GetUpVector	Gets the camera's up direction.
 GetViewVector	Gets the direction in which the camera is looking.
 SetPositionCartesian	Sets the Cartesian coordinates for the camera position.
 SetPositionEntity	Sets the entity to which the camera is attached.
 SetPositionSpherical	Sets the spherical coordinates for the camera position relative to the target, based on a sphere around the Z axis with the zero (0) point at the target.
 SetTargetPointEntity	Gets the target point on the entity for the camera.

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See Also

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ICamera Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : ICamera Interface

For a list of all members of this type, see [ICamera members](#).

Public Properties

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 AspectRatio	Gets the aspect ratio (width/height) of the field of view rectangle for the camera. NOTE: This property is a get-only property. Set is not implemented.

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 DepthOfFieldOffset	Gets the approximate distance from the camera's focal plane to point where focus is lost. NOTE: This property is a get-only property. Set is not implemented.
 FieldOfViewAngle	Gets the camera's horizontal angle of the field of view. NOTE: This property is a get-only property. Set is not implemented.
 FieldOfViewDepth	Gets the camera's depth of the field of view. NOTE: This property is a get-only property. Set is not implemented.
 FieldOfViewHeight	Gets the camera's height of the field of view. NOTE: This property is a get-only property. Set is not implemented.
 ID	Gets the ID for this camera.
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 Pitch	Gets or sets the pitch (up or down angle) of a floating camera .
 PositionType	Gets or sets the camera position type.
 RotationRollAngle	Gets or sets the camera's roll angle.
 RotationRollBySelection	Gets or sets whether you can select the camera's rotation roll direction.
 RotationRollEntity	Gets or sets the entity (line, edge, face, or plane) that defines the camera's rotation up direction.
 RotationRollFlipDirection	Gets or sets whether to flip the direction of the camera 180.
 TargetPointBySelection	Gets or sets whether you can set the target point.
 TargetPointPosition	Gets or sets the position of the target point.
 Type	Gets or sets the type of camera.
 Yaw	Gets or sets the yaw (side-to-side angle) of a floating camera .

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 See Also

[ICamera Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

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AspectRatio Property (ICamera)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICamera Interface](#) : AspectRatio Property (ICamera)

Gets the aspect ratio (width/height) of the field of view rectangle for the camera.

NOTE: This property is a get-only property. Set is not implemented.

 .NET Syntax

Visual Basic (Declaration)

Property AspectRatio As System.Double

Visual Basic (Usage)



```
Dim instance As ICamera
Dim value As System.Double
```

```
instance.AspectRatio = value  
value = instance.AspectRatio
```

C#

```
System.double AspectRatio {get; set;}
```

C++/CLI

```
property System.double AspectRatio {  
    System.double get();  
    void set ( & System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Width/height ratio of the view rectangle for the camera

Visual Basic for Applications (VBA) Syntax

See [Camera::AspectRatio](#).

Remarks

The rectangle is best-fit into a rendering window to determine the actual composition.

See Also

[ICamera Interface](#)
[ICamera Members](#)

Availability

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DepthOfFieldEnabled Property (ICamera)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICamera Interface](#) : DepthOfFieldEnabled Property (ICamera)

Gets whether depth of field effects are enabled or disabled.

NOTE: This property is a get-only property. Set is not implemented.

.NET Syntax

Visual Basic (Declaration)

```
Property DepthOfFieldEnabled As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ICamera  
Dim value As System.Boolean  
  
instance.DepthOfFieldEnabled = value  
value = instance.DepthOfFieldEnabled
```

C#

```
System.bool DepthOfFieldEnabled {get; set;}
```

C++/CLI

```
property System.bool DepthOfFieldEnabled {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True if depth of field effects are enabled, false if they are disabled

Visual Basic for Applications (VBA) Syntax

See [Camera::DepthOfFieldEnabled](#).

Example

[Insert Camera \(C#\)](#)
[Insert Camera \(VB.NET\)](#)
[Insert Camera \(VBA\)](#)

See Also

[ICamera Interface](#)
[ICamera Members](#)
[ICamera::DepthOfFieldOffset Property](#)

Availability

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DepthOfFieldOffset Property (ICamera)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICamera Interface](#) : DepthOfFieldOffset Property (ICamera)

Gets the approximate distance from the camera's focal plane to point where focus is lost.

NOTE: This property is a get-only property. Set is not implemented.

.NET Syntax

Visual Basic (Declaration)

Property DepthOfFieldOffset As System.Double

Visual Basic (Usage)



```
Dim instance As ICamera
Dim value As System.Double

instance.DepthOfFieldOffset = value
value = instance.DepthOfFieldOffset
```

C#

```
System.double DepthOfFieldOffset {get; set;}
```

C++/CLI

```
property System.double DepthOfFieldOffset {
    System.double get();
    void set ( & System.double value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Approximate distance from the camera's focal plane to the point where focus is lost

Visual Basic for Applications (VBA) Syntax

See [Camera::DepthOfFieldOffset](#).

Example

[Insert Camera \(C#\)](#)
[Insert Camera \(VB.NET\)](#)
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See Also

[ICamera Interface](#)
[ICamera Members](#)
[ICamera::DepthOfFieldEnabled Property](#)

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FieldOfViewAngle Property (ICamera)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICamera Interface](#) : FieldOfViewAngle Property (ICamera)

Gets the camera's horizontal angle of the field of view.

NOTE: This property is a get-only property. Set is not implemented.

.NET Syntax

Visual Basic (Declaration)

```
Property FieldOfViewAngle As System.Double
```

Visual Basic (Usage)



```
Dim instance As ICamera
Dim value As System.Double

instance.FieldOfViewAngle = value

value = instance.FieldOfViewAngle
```

C#

```
System.double FieldOfViewAngle {get; set;}
```

C++/CLI

```
property System.double FieldOfViewAngle {
    System.double get();
    void set ( & System.double value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Camera's horizontal angle of the field of view in radians

Visual Basic for Applications (VBA) Syntax

See [Camera::FieldOfViewAngle](#).

Example

[Insert Camera \(VBA\)](#)
[Insert Camera \(C#\)](#)
[Insert Camera \(VB.NET\)](#)

Remarks

This property is only meaningful for a perspective camera.

See Also

[ICamera Interface](#)
[ICamera Members](#)
[ICamera::FieldOfViewDepth Property](#)
[ICamera::FieldOfViewHeight Property](#)
[ICamera::Perspective Property](#)

Availability

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FieldOfViewDepth Property (ICamera)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICamera Interface](#) : FieldOfViewDepth Property (ICamera)

Gets the camera's depth of the field of view.

NOTE: This property is a get-only property. Set is not implemented.

.NET Syntax

Visual Basic (Declaration)

```
Property FieldOfViewDepth As System.Double
```

Visual Basic (Usage)



```
Dim instance As ICamera  
Dim value As System.Double  
  
instance.FieldOfViewDepth = value  
  
value = instance.FieldOfViewDepth
```

C#

```
System.double FieldOfViewDepth {get; set;}
```

C++/CLI

```
property System.double FieldOfViewDepth {  
    System.double get();  
    void set ( & System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Camera's depth of the field of view

Visual Basic for Applications (VBA) Syntax

See [Camera::FieldOfViewDepth](#).

Example

[Insert Camera \(C#\)](#)
[Insert Camera \(VB.NET\)](#)
[Insert Camera \(VBA\)](#)

Remarks

This property is only meaningful for a perspective camera.

See Also

[ICamera Interface](#)
[ICamera Members](#)
[ICamera::FieldOfViewAngle Property](#)
[ICamera::FieldOfViewHeight Property](#)
[ICamera::Perspective Property](#)

Availability

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FieldOfViewHeight Property (ICamera)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICamera Interface](#) : FieldOfViewHeight Property (ICamera)

Gets the camera's height of the field of view.

NOTE: This property is a get-only property. Set is not implemented.

.NET Syntax

Visual Basic (Declaration)

```
Property FieldOfViewHeight As System.Double
```

Visual Basic (Usage)



```
Dim instance As ICamera  
Dim value As System.Double  
  
instance.FieldOfViewHeight = value  
  
value = instance.FieldOfViewHeight
```

C#

```
System.double FieldOfViewHeight {get; set;}
```

C++/CLI

```
property System.double FieldOfViewHeight {  
    System.double get();  
    void set ( & System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Camera's height of the field of view

Visual Basic for Applications (VBA) Syntax

See [Camera::FieldOfViewHeight](#).

Example

[Insert Camera \(C#\)](#)
[Insert Camera \(VB.NET\)](#)
[Insert Camera \(VBA\)](#)

Remarks

This property is only meaningful for a perspective camera.

See Also

[ICamera Interface](#)
[ICamera Members](#)
[ICamera::FieldOfViewAngle Property](#)
[ICamera::FieldOfViewDepth Property](#)
[ICamera::Perspective Property](#)

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GetFocalDistance Method (ICamera)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICamera Interface](#) : GetFocalDistance Method (ICamera)

Gets the camera's focal distance.

.NET Syntax

Visual Basic (Declaration)

```
Function GetFocalDistance() As System.Double
```

Visual Basic (Usage)



```
Dim instance As ICamera  
Dim value As System.Double  
  
value = instance.GetFocalDistance()
```

C#

```
System.double GetFocalDistance()
```

C++/CLI

```
System.double GetFocalDistance();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Camera's focal distance

Visual Basic for Applications (VBA) Syntax

See [Camera::GetFocalDistance](#).

Example

[Insert Camera \(C#\)](#)
[Insert Camera \(VB.NET\)](#)
[Insert Camera \(VBA\)](#)

Remarks

If the camera is locked to a reference, then the return value might be different than the numeric value shown in the Camera PropertyManager.

See Also

[ICamera Interface](#)
[ICamera Members](#)

Availability

GetPointOfInterestDistance Method (ICamera)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICamera Interface](#) : GetPointOfInterestDistance Method (ICamera)

Gets the distance of the point of interest from the camera.

.NET Syntax

Visual Basic (Declaration)

```
Function GetPointOfInterestDistance() As System.Double
```

Visual Basic (Usage)



```
Dim instance As ICamera  
Dim value As System.Double  
  
value = instance.GetPointOfInterestDistance()
```

C#

```
System.double GetPointOfInterestDistance()
```

C++/CLI

```
System.double GetPointOfInterestDistance();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Distance of the point of interest from the camera

Visual Basic for Applications (VBA) Syntax

See [Camera::GetPointOfInterestDistance](#).

See Also

[ICamera Interface](#)
[ICamera Members](#)

Availability

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GetPosition Method (ICamera)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICamera Interface](#) : GetPosition Method (ICamera)

Gets the current position of the camera.

.NET Syntax

Visual Basic (Declaration)

```
Function GetPosition() As MathPoint
```

Visual Basic (Usage)



```
Dim instance As ICamera  
Dim value As MathPoint
```

```
value = instance.GetPosition()
```

C#

```
MathPoint GetPosition()
```

C++/CLI

```
MathPoint^ GetPosition();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to [IMathPoint](#) object

Visual Basic for Applications (VBA) Syntax

See [Camera::GetPosition](#).

See Also

[ICamera Interface](#)
[ICamera Members](#)
[ICamera::GetPositionCartesian Method](#)
[ICamera::GetPositionEntity Method](#)
[ICamera::GetPositionSpherical Method](#)
[ICamera::PositionType Property](#)
[ICamera::LockCameraPosition Property](#)

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GetPositionCartesian Method (ICamera)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICamera Interface](#) : GetPositionCartesian Method (ICamera)

Gets the Cartesian coordinates for the camera position.

.NET Syntax

Visual Basic (Declaration)

```
Sub GetPositionCartesian( _  
    ByRef X As System.Double, _  
    ByRef Y As System.Double, _  
    ByRef Z As System.Double _  
)
```

Visual Basic (Usage)



```
Dim instance As ICamera  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double
```

```
instance.GetPositionCartesian(X, Y, Z)
```

C#

```
void GetPositionCartesian(  
    out System.double X,  
    out System.double Y,  
    out System.double Z  
)
```

C++/CLI

```

void GetPositionCartesian(
& [Out] System.double X,
& [Out] System.double Y,
& [Out] System.double Z
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X
x coordinate relative to the model origin

Y
y coordinate relative to the model origin

Z
z coordinate relative to the model origin

Visual Basic for Applications (VBA) Syntax

See [Camera::GetPositionCartesian](#).

See Also

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[ICamera Members](#)
[ICamera::GetPositionCartesian Method](#)
[ICamera::GetPositionEntity Method](#)
[ICamera::GetPositionSpherical Method](#)
[ICamera::PositionType Property](#)
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[ICamera::SetPositionEntity Method](#)
[ICamera::SetPositionSpherical Method](#)

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GetPositionEntity Method (ICamera)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICamera Interface](#) : GetPositionEntity Method (ICamera)

Gets the entity to which the camera is attached.

.NET Syntax

Visual Basic (Declaration)

```

Function GetPositionEntity( _
    ByRef Point As MathPoint, _
    ByRef PercentPosition As System.Double _
) As System.Object

```

Visual Basic (Usage)



```

Dim instance As ICamera
Dim Point As MathPoint
Dim PercentPosition As System.Double
Dim value As System.Object

value = instance.GetPositionEntity(Point, PercentPosition)

```

C#

```

System.object GetPositionEntity(
    out MathPoint Point,
    out System.double PercentPosition
)

```

C++/CLI

```
System.Object^ GetPositionEntity(
& [Out] MathPoint^ Point,
& [Out] System.double PercentPosition
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Point
[IMathPoint](#) object indicating the camera position
PercentPosition
Camera point distance along the entity

Return Value

Entity for the camera position

Visual Basic for Applications (VBA) Syntax

See [Camera::GetPositionEntity](#).

See Also

[ICamera Interface](#)
[ICamera Members](#)
[ICamera::GetPosition Method](#)
[ICamera::GetPositionCartesian Method](#)
[ICamera::GetPositionSpherical Method](#)
[ICamera::LockCameraPosition Property](#)
[ICamera::PositionType Property](#)
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[ICamera::SetPositionEntity Method](#)
[ICamera::SetPositionSpherical Method](#)

Availability

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GetPositionSpherical Method (ICamera)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICamera Interface](#) : GetPositionSpherical Method (ICamera)

Gets the spherical coordinates for the camera position relative to the target, based on a sphere around the Z axis with the zero (0) point at the target.

.NET Syntax

Visual Basic (Declaration)

```
Sub GetPositionSpherical( _
    ByRef L As System.Double, _
    ByRef A1 As System.Double, _
    ByRef A2 As System.Double _
)
```

Visual Basic (Usage)

```
Dim instance As ICamera
Dim L As System.Double
Dim A1 As System.Double
Dim A2 As System.Double

instance.GetPositionSpherical(L, A1, A2)
```

C#

```
void GetPositionSpherical(
    out System.double L,
    out System.double A1,
    out System.double A2
)
```



```
void GetPositionSpherical(
& [Out] System.double L,
& [Out] System.double A1,
& [Out] System.double A2
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

L
Distance from target

A1
Longitude about target

A2
Latitude about target

Visual Basic for Applications (VBA) Syntax

See [Camera::GetPositionSpherical](#).

See Also

[ICamera Interface](#)
[ICamera Members](#)
[ICamera::GetPosition Method](#)
[ICamera::GetPositionCartesian Method](#)
[ICamera::GetPositionEntity Method](#)
[ICamera::GetPositionSpherical Method](#)
[ICamera::SetPositionCartesian Method](#)
[ICamera::SetPositionEntity Method](#)
[ICamera::SetPositionSpherical Method](#)
[ICamera::LockCameraPosition Property](#)
[ICamera::PositionType Property](#)

Availability

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GetTargetPointEntity Method (ICamera)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICamera Interface](#) : GetTargetPointEntity Method (ICamera)

Gets the target point on the entity for the camera.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTargetPointEntity( _
    ByRef Point As MathPoint, _
    ByRef PercentTarget As System.Double _
) As System.Object
```

Visual Basic (Usage)

```
Dim instance As ICamera
Dim Point As MathPoint
Dim PercentTarget As System.Double
Dim value As System.Object

value = instance.GetTargetPointEntity(Point, PercentTarget)
```

C#

```
System.object GetTargetPointEntity(
    out MathPoint Point,
    out System.double PercentTarget
)
```



```
System.Object^ GetTargetPointEntity(
& [Out] MathPoint^ Point,
& [Out] System.double PercentTarget
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Point
[IMathPoint](#) object indicating the target point for the camera
PercentTarget
 Target point distance along the entity

Return Value

Entity for the target point

Visual Basic for Applications (VBA) Syntax

See [Camera::GetTargetPointEntity](#).

See Also

[ICamera Interface](#)
[ICamera Members](#)
[ICamera::SetTargetPointEntity Method](#)
[ICamera::TargetPointBySelection Property](#)
[ICamera::TargetPointPosition Property](#)

Availability

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GetUpVector Method (ICamera)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICamera Interface](#) : GetUpVector Method (ICamera)

Gets the camera's up direction.

.NET Syntax

Visual Basic (Declaration)

```
Function GetUpVector() As MathVector
```

Visual Basic (Usage)



```
Dim instance As ICamera
Dim value As MathVector

value = instance.GetUpVector()
```

C#

```
MathVector GetUpVector()
```

C++/CLI

```
MathVector^ GetUpVector();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Point to [IMathVector](#) object

Visual Basic for Applications (VBA) Syntax

See [Camera::GetUpVector](#).

See Also

[ICamera Interface](#)
[ICamera Members](#)
[ICamera::GetViewVector Method](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetViewVector Method (ICamera)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICamera Interface](#) : GetViewVector Method (ICamera)

Gets the direction in which the camera is looking.

.NET Syntax

Visual Basic (Declaration)

```
Function GetViewVector() As MathVector
```

Visual Basic (Usage)



```
Dim instance As ICamera  
Dim value As MathVector  
  
value = instance.GetViewVector()
```

C#

```
MathVector GetViewVector()
```

C++/CLI

```
MathVector^ GetViewVector();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to [IMathVector](#) object

Visual Basic for Applications (VBA) Syntax

See [Camera::GetViewVector](#).

Remarks

Most of the time, the return value is the vector from the camera position to the target, normalized.

See Also

[ICamera Interface](#)
[ICamera Members](#)
[ICamera::GetUpVector Method](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14

SOLIDWORKS API Help

[Send comments on this topic.](#)

ID Property (ICamera)

[See Also](#) [Example](#)



Gets the ID for this camera.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property ID As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ICamera  
Dim value As System.Integer  
  
value = instance.ID
```

C#

```
System.int ID {get;}
```

C++/CLI

```
property System.int ID {  
    System.int get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

ID for camera

Visual Basic for Applications (VBA) Syntax

See [Camera::ID](#).

Example

[Turn Cameras On and Off \(VBA\)](#)
[Insert Camera \(C#\)](#)
[Insert Camera \(VB.NET\)](#)
[Insert Camera \(VBA\)](#)

See Also

[ICamera Interface](#)
[ICamera Members](#)
[IModelDocExtension::GetCameraById Method](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14

LockCameraPosition Property (ICamera)

[See Also](#)



Gets or sets whether the camera position is locked.

.NET Syntax

Visual Basic (Declaration)

```
Property LockCameraPosition As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ICamera
Dim value As System.Boolean

instance.LockCameraPosition = value
value = instance.LockCameraPosition
```

C#

```
System.bool LockCameraPosition {get; set;}
```

C++/CLI

```
property System.bool LockCameraPosition {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True if camera position is locked, false if not

Visual Basic for Applications (VBA) Syntax

See [Camera::LockCameraPosition](#).

See Also

[ICamera Interface](#)
[ICamera Members](#)
[ICamera::GetPosition Method](#)
[ICamera::GetPositionCartesian Method](#)
[ICamera::GetPositionEntity Method](#)
[ICamera::GetPositionSpherical Method](#)
[ICamera::SetPositionCartesian Method](#)
[ICamera::SetPositionEntity Method](#)
[ICamera::SetPositionSpherical Method](#)
[ICamera::PositionType Property](#)

Availability

SOLIDWORKS 2007 SP1, Revision Number 15

SOLIDWORKS API Help

[Send comments on this topic.](#)

Perspective Property (ICamera)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICamera Interface](#) : Perspective Property (ICamera)

Gets whether the camera is in perspective mode or not.

NOTE: This property is a get-only property. Set is not implemented.

.NET Syntax

Visual Basic (Declaration)

```
Property Perspective As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ICamera
Dim value As System.Boolean

instance.Perspective = value
value = instance.Perspective
```

C#

```
System.bool Perspective {get; set;}
```

C++/CLI

```
property System.bool Perspective {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True if camera is in perspective mode, false if not

Visual Basic for Applications (VBA) Syntax

See [Camera::Perspective](#).

Example

[Insert Camera \(C#\)](#)
[Insert Camera \(VB.NET\)](#)
[Insert Camera \(VBA\)](#)

See Also

[ICamera Interface](#)
[ICamera Members](#)
[ICamera::FieldOfViewAngle Property](#)
[ICamera::FieldOfViewDepth Property](#)
[ICamera::FieldOfViewHeight Property](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14

SOLIDWORKS API Help

[Send comments](#) on this topic.

Pitch Property (ICamera)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICamera Interface](#) : Pitch Property (ICamera)

Gets or sets the pitch (up or down angle) of a [floating camera](#).

.NET Syntax

Visual Basic (Declaration)

```
Property Pitch As System.Double
```

Visual Basic (Usage)



```
Dim instance As ICamera  
Dim value As System.Double  
  
instance.Pitch = value  
  
value = instance.Pitch
```

C#

```
System.double Pitch {get; set;}
```

C++/CLI

```
property System.double Pitch {  
    System.double get();  
    void set ( & System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Pitch

Visual Basic for Applications (VBA) Syntax

See [Camera::Pitch](#).

Example

[Insert and Rotate Camera \(C#\)](#)
[Insert and Rotate Camera \(VB.NET\)](#)
[Insert and Rotate Camera \(VBA\)](#)

See Also

[ICamera Interface](#)
[ICamera Members](#)

Availability

SOLIDWORKS 2009 SP4, Revision Number 17.4

SOLIDWORKS API Help

[Send comments on this topic.](#)

PositionType Property (ICamera)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICamera Interface](#) : PositionType Property (ICamera)

Gets or sets the camera position type.

.NET Syntax

Visual Basic (Declaration)

Property PositionType As System.Integer

Visual Basic (Usage)



```
Dim instance As ICamera  
Dim value As System.Integer  
  
instance.PositionType = value  
value = instance.PositionType
```

C#

```
System.int PositionType {get; set;}
```

C++/CLI

```
property System.int PositionType {  
    System.int get();  
    void set ( & System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Camera position type as defined in [swCameraPositionType_e](#).

Visual Basic for Applications (VBA) Syntax

See [Camera::PositionType](#).

Example

See Also

[ICamera Interface](#)
[ICamera Members](#)
[ICamera::GetPosition Method](#)
[ICamera::GetPositionCartesian Method](#)
[ICamera::GetPositionEntity Method](#)
[ICamera::GetPositionSpherical Method](#)
[ICamera::SetPositionCartesian Method](#)
[ICamera::SetPositionEntity Method](#)
[ICamera::SetPositionSpherical Method](#)
[ICamera::LockCameraPosition Property](#)

Availability

SOLIDWORKS 2007 SP1, Revision Number 14

SOLIDWORKS API Help

[Send comments on this topic.](#)

RotationRollAngle Property (ICamera)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICamera Interface](#) : RotationRollAngle Property (ICamera)

Gets or sets the camera's roll angle.

.NET Syntax

Visual Basic (Declaration)

Property RotationRollAngle As System.Double

Visual Basic (Usage)



```
Dim instance As ICamera
Dim value As System.Double

instance.RotationRollAngle = value
value = instance.RotationRollAngle
```

C#

```
System.double RotationRollAngle {get; set;}
```

C++/CLI

```
property System.double RotationRollAngle {
    System.double get();
    void set ( & System.double value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Roll angle for camera

Visual Basic for Applications (VBA) Syntax

See [Camera::RotationRollAngle](#).

See Also

[ICamera Interface](#)
[ICamera Members](#)
[ICamera::RotationRollBySelection Property](#)
[ICamera::RotationRollEntity Property](#)
[ICamera::RotationRollFlipDirection Property](#)

Availability

SOLIDWORKS 2007 SP1, Revision Number 15

RotationRollBySelection Property (ICamera)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICamera Interface](#) : RotationRollBySelection Property (ICamera)

Gets or sets whether you can select the camera's rotation roll direction.

.NET Syntax

Visual Basic (Declaration)

Property RotationRollBySelection As System.Boolean

Visual Basic (Usage)



```
Dim instance As ICamera
Dim value As System.Boolean

instance.RotationRollBySelection = value
value = instance.RotationRollBySelection
```

C#

```
System.bool RotationRollBySelection {get; set;}
```

C++/CLI

```
property System.bool RotationRollBySelection {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True if you can select the camera's rotation roll, false if not

Visual Basic for Applications (VBA) Syntax

See [Camera::RotationRollBySelection](#).

See Also

[ICamera Interface](#)
[ICamera Members](#)
[ICamera::RotationRollAngle Property](#)
[ICamera::RotationRollEntity Property](#)
[ICamera::RotationRollFlipDirection Property](#)

Availability

SOLIDWORKS 2007 SP1, Revision Number 15

RotationRollEntity Property (ICamera)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICamera Interface](#) : RotationRollEntity Property (ICamera)

Gets or sets the entity (line, edge, face, or plane) that defines the camera's rotation up direction.

.NET Syntax

Visual Basic (Declaration)

Property RotationRollEntity As System.Object

Visual Basic (Usage)



```
Dim instance As ICamera
Dim value As System.Object

instance.RotationRollEntity = value
value = instance.RotationRollEntity
```

C#

```
System.object RotationRollEntity {get; set;}
```

C++/CLI

```
property System.Object^ RotationRollEntity {
    System.Object^ get();
    void set ( &    System.Object^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Pointer to the entity (line, edge, face, or plane) that defines the camera's rotation up direction

Visual Basic for Applications (VBA) Syntax

See [Camera::RotationRollEntity](#).

See Also

[ICamera Interface](#)
[ICamera Members](#)
[ICamera::RotationRollAngle Property](#)
[ICamera::RotationRollBySelection Property](#)
[ICamera::RotationRollFlipDirection Property](#)

Availability

SOLIDWORKS 2007 SP1, Revision Number 15

SOLIDWORKS API Help

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RotationRollFlipDirection Property (ICamera)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICamera Interface](#) : RotationRollFlipDirection Property (ICamera)

Gets or sets whether to flip the direction of the camera 180.

.NET Syntax

Visual Basic (Declaration)

```
Property RotationRollFlipDirection As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ICamera
Dim value As System.Boolean

instance.RotationRollFlipDirection = value
value = instance.RotationRollFlipDirection
```

C#

```
System.bool RotationRollFlipDirection {get; set;}
```

C++/CLI

```
property System.bool RotationRollFlipDirection {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to flip the direction of the camera 180, false to not

Visual Basic for Applications (VBA) Syntax

See [Camera::RotationRollFlipDirection](#).

See Also

[ICamera Interface](#)
[ICamera Members](#)
[ICamera::RotationRollAngle Property](#)
[ICamera::RotationRollBySelection Property](#)
[ICamera::RotationRollEntity Property](#)

Availability

SOLIDWORKS 2007 SP1, Revision Number 15

SOLIDWORKS API Help

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SetPositionCartesian Method (ICamera)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICamera Interface](#) : SetPositionCartesian Method (ICamera)

Sets the Cartesian coordinates for the camera position.

.NET Syntax

Visual Basic (Declaration)

```
Sub SetPositionCartesian( _
    ByVal X As System.Double, _
    ByVal Y As System.Double, _
    ByVal Z As System.Double _
)
```

Visual Basic (Usage)



```
Dim instance As ICamera
Dim X As System.Double
Dim Y As System.Double
Dim Z As System.Double
```

```
instance.SetPositionCartesian(X, Y, Z)
```

C#

```
void SetPositionCartesian(
    System.double X,
    System.double Y,
    System.double Z
)
```

C++/CLI

```
void SetPositionCartesian(
    & System.double X,
    & System.double Y,
    & System.double Z
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X
x coordinate relative to the model origin
Y
y coordinate relative to the model origin
Z
z coordinate relative to the model origin

Visual Basic for Applications (VBA) Syntax

See [Camera::SetPositionCartesian](#).

See Also

[ICamera Interface](#)
[ICamera Members](#)
[ICamera::GetPosition Method](#)
[ICamera::GetPositionCartesian Method](#)
[ICamera::GetPositionEntity Method](#)
[ICamera::GetPositionSpherical Method](#)
[ICamera::SetPositionEntity Method](#)
[ICamera::SetPositionSpherical Method](#)
[ICamera::PositionType Property](#)
[ICamera::LockCameraPosition Property](#)

Availability

SOLIDWORKS 2007 SP1, Revision Number 15

SOLIDWORKS API Help

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SetPositionEntity Method (ICamera)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICamera Interface](#) : SetPositionEntity Method (ICamera)

Sets the entity to which the camera is attached.

.NET Syntax

Visual Basic (Declaration)

```
Sub SetPositionEntity( _  
    ByVal Point As MathPoint, _  
    ByVal PercentPosition As System.Double, _  
    ByVal PositionEntity As System.Object _  
)
```

Visual Basic (Usage)

```
Dim instance As ICamera  
Dim Point As MathPoint  
Dim PercentPosition As System.Double  
Dim PositionEntity As System.Object  
  
instance.SetPositionEntity(Point, PercentPosition, PositionEntity)
```

C#

```
void SetPositionEntity(  
    MathPoint Point,  
    System.double PercentPosition,  
    System.object PositionEntity  
)
```

C++/CLI

```
void SetPositionEntity(  
& MathPoint^ Point,  
& System.double PercentPosition,  
& System.Object^ PositionEntity  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Point

Pointer to the [IMathPoint](#) object indicating the camera position

PercentPosition

Camera point distance along the entity

PositionEntity

Entity for the camera position

Visual Basic for Applications (VBA) Syntax

See [Camera::SetPositionEntity](#).

See Also

[ICamera Interface](#)

[ICamera Members](#)

[ICamera::GetPosition Method](#)

[ICamera::GetPositionCartesian Method](#)

[ICamera::GetPositionEntity Method](#)

[ICamera::GetPositionSpherical Method](#)

[ICamera::SetPositionCartesian Method](#)

[ICamera::SetPositionSpherical Method](#)

[ICamera::LockCameraPosition Property](#)

[ICamera::PositionType Property](#)

Availability

SOLIDWORKS 2007 SP1, Revision Number 15

SOLIDWORKS API Help

[Send comments on this topic](#).

SetPositionSpherical Method (ICamera)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICamera Interface](#) : SetPositionSpherical Method (ICamera)

Sets the spherical coordinates for the camera position relative to the target, based on a sphere around the Z axis with the zero (0) point at the target.

.NET Syntax

Visual Basic (Declaration)

```
Sub SetPositionSpherical( _  
    ByVal L As System.Double, _  
    ByVal A1 As System.Double, _  
    ByVal A2 As System.Double _  
)
```

Visual Basic (Usage)



```
Dim instance As ICamera  
Dim L As System.Double  
Dim A1 As System.Double  
Dim A2 As System.Double
```

```
instance.SetPositionSpherical(L, A1, A2)
```

C#

```
void SetPositionSpherical(  
    System.double L,  
    System.double A1,  
    System.double A2  
)
```

C++/CLI

```

void SetPositionSpherical(
& System.double L,
& System.double A1,
& System.double A2
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

L
Distance from target
A1
Longitude about target
A2
Latitude about target

Return Value

Latitude about target

Visual Basic for Applications (VBA) Syntax

See [Camera::SetPositionSpherical](#).

See Also

[ICamera Interface](#)
[ICamera Members](#)
[ICamera::GetPosition Method](#)
[ICamera::GetPositionCartesian Method](#)
[ICamera::GetPositionEntity Method](#)
[ICamera::GetPositionSpherical Method](#)
[ICamera::SetPositionCartesian Method](#)
[ICamera::SetPositionEntity Method](#)
[ICamera::LockCameraPosition Property](#)
[ICamera::ICamera::PositionType Property](#)

Availability

SOLIDWORKS 2007 SP1, Revision Number 15

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetTargetPointEntity Method (ICamera)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICamera Interface](#) : SetTargetPointEntity Method (ICamera)

Gets the target point on the entity for the camera.

.NET Syntax

Visual Basic (Declaration)

```

Sub SetTargetPointEntity( _
    ByVal Point As MathPoint, _
    ByVal PercentTarget As System.Double, _
    ByVal TargetPointEntity As System.Object _
)

```

Visual Basic (Usage)



```

Dim instance As ICamera
Dim Point As MathPoint
Dim PercentTarget As System.Double
Dim TargetPointEntity As System.Object

instance.SetTargetPointEntity(Point, PercentTarget, TargetPointEntity)

```

C#

```

void SetTargetPointEntity(
    MathPoint Point,
    System.double PercentTarget,

```

```
System.Object TargetPointEntity
)
```

C++/CLI

```
void SetTargetPointEntity(
& MathPoint^ Point,
& System.Double PercentTarget,
& System.Object^ TargetPointEntity
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Point
Pointer to the [IMathPoint](#) object indicating the target point for the camera
PercentTarget
Target point distance along the entity
TargetPointEntity
Entity for the target point

Visual Basic for Applications (VBA) Syntax

See [Camera::SetTargetPointEntity](#).

See Also

[ICamera Interface](#)
[ICamera Members](#)
[ICamera::GetTargetPointEntity Method](#)
[ICamera::TargetPointBySelection Property](#)
[ICamera::TargetPointPosition Property](#)

Availability

SOLIDWORKS 2007 SP1, Revision Number 15

SOLIDWORKS API Help

[Send comments on this topic.](#)

TargetPointBySelection Property (ICamera)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICamera Interface](#) : TargetPointBySelection Property (ICamera)

Gets or sets whether you can set the target point.

.NET Syntax

Visual Basic (Declaration)

```
Property TargetPointBySelection As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ICamera
Dim value As System.Boolean

instance.TargetPointBySelection = value
value = instance.TargetPointBySelection
```

C#

```
System.bool TargetPointBySelection {get; set;}
```

C++/CLI

```
property System.bool TargetPointBySelection {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True if you can set the target point, false if not

Visual Basic for Applications (VBA) Syntax

See [Camera::TargetPointBySelection](#).

See Also

[ICamera Interface](#)
[ICamera Members](#)
[ICamera::TargetPointPosition Property](#)
[ICamera::GetTargetPointEntity Method](#)
[ICamera::SetTargetPointEntity Method](#)

Availability

SOLIDWORKS 2007 SP1, Revision Number 15

SOLIDWORKS API Help

[Send comments on this topic.](#)

TargetPointPosition Property (ICamera)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICamera Interface](#) : TargetPointPosition Property (ICamera)

Gets or sets the position of the target point.

.NET Syntax

Visual Basic (Declaration)

Property TargetPointPosition As MathPoint

Visual Basic (Usage)



```
Dim instance As ICamera
Dim value As MathPoint

instance.TargetPointPosition = value

value = instance.TargetPointPosition
```

C#

```
MathPoint TargetPointPosition {get; set;}
```

C++/CLI

```
property MathPoint^ TargetPointPosition {
    MathPoint^ get();
    void set ( & MathPoint^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Position of target point

Visual Basic for Applications (VBA) Syntax

See [Camera::TargetPointPosition](#).

See Also

[ICamera Interface](#)
[ICamera Members](#)
[ICamera::GetTargetPointEntity Method](#)
[ICamera::SetTargetPointEntity Method](#)
[ICamera::TargetPointBySelection Property](#)

Availability

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Type Property (ICamera)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICamera Interface](#) : Type Property (ICamera)

Gets or sets the type of camera.

Visual Basic Syntax

Visual Basic (Declaration)

Property Type As System.Integer

Visual Basic (Usage)



```
Dim instance As ICamera
Dim value As System.Integer

instance.Type = value
value = instance.Type
```

C#

```
System.int Type {get; set;}
```

C++/CLI

```
property System.int Type {
    System.int get();
    void set ( & System.int value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Type of camera as defined in [swCameraType_e](#)

Visual Basic for Applications (VBA) Syntax

See [Camera::Type](#).

Example

[Turn Cameras On and Off \(VBA\)](#)
[Turn Cameras On and Off \(VB.NET\)](#)
[Turn Cameras On and Off \(C#\)](#)

See Also

[ICamera Interface](#)
[ICamera Members](#)
[ICamera::TargetPointPosition Property](#)
[ICamera::TargetPointBySelection Property](#)
[ICamera::RotationRollFlipDirection Property](#)
[ICamera::SetTargetPointEntity Method](#)
[ICamera::GetTargetPointEntity Method](#)

Availability

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SOLIDWORKS API Help

[Send comments on this topic.](#)

Yaw Property (ICamera)

[See Also](#) [Example](#)



Gets or sets the yaw (side-to-side angle) of a [floating camera](#).

.NET Syntax

Visual Basic (Declaration)

Property Yaw As System.Double

Visual Basic (Usage)



```
Dim instance As ICamera  
Dim value As System.Double
```

```
instance.Yaw = value
```

```
value = instance.Yaw
```

C#

```
System.double Yaw {get; set;}
```

C++/CLI

```
property System.double Yaw {  
    System.double get();  
    void set ( & System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Yaw

Visual Basic for Applications (VBA) Syntax

See [Camera::Yaw](#).

Example

[Insert and Rotate Camera \(C#\)](#)
[Insert and Rotate Camera \(VB.NET\)](#)
[Insert and Rotate Camera \(VBA\)](#)

See Also

[ICamera Interface](#)
[ICamera Members](#)

Availability

SOLIDWORKS 2009 SP4, Revision Number 17.4

ICamFollowerMateFeatureData (Type)

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ICamFollowerMateFeatureData Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : ICamFollowerMateFeatureData Interface

Allows access to a cam-follower mate feature.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface ICamFollowerMateFeatureData
```

Visual Basic (Usage)



```
Dim instance As ICamFollowerMateFeatureData
```

C#

```
public interface ICamFollowerMateFeatureData
```

C++/CLI

```
public interface class ICamFollowerMateFeatureData
```

Visual Basic for Applications (VBA) Syntax

See [CamFollowerMateFeatureData](#).

Example

```
'VBA
'*****
' 1. Open public_documents\samples\tutorial\api\MechanicalMates\Cam-Follower.sldasm.
' 2. Delete CamMateCoincident1 from the Mates folder in the FeatureManager design tree.
' 3. Run the macro.
' 4. Inspect the Mates folder in the FeatureManager design tree.
' *****
Dim swApp As SldWorks.SldWorks
Dim Part As SldWorks.ModelDoc2
Dim boolstatus As Boolean

Option Explicit

Sub main()

    Set swApp = Application.SldWorks
    Set Part = swApp.ActiveDoc

    boolstatus = Part.Extension.SelectByRay(-2.29201081452857E-03, 7.61655216381314E-02, 9.81125242168446E-03, -0.577381545199981, -0.577287712085548, -0.577381545199979, 2.88013169157359E-03, 2, True, 1, 0)
    boolstatus = Part.Extension.SelectByRay(0, 0.0762, 0, -0.577381545199981, -0.577287712085548, -0.577381545199979, 2.88013169157359E-03, 3, True, 8, 0)

    ' Create CamFollowerMateFeatureData
    Dim MateData As SldWorks.CamFollowerMateFeatureData
    Set MateData = Part.CreateMateData(9)

    ' Set the Entities To Mate
    Dim FirstEntityToMate As Object
    Dim SecondEntityToMate As Object
    Set FirstEntityToMate = Part.SelectionManager.GetSelectedObject6(1, -1)
    MateData.EntitiesToMate(0) = FirstEntityToMate
    Set SecondEntityToMate = Part.SelectionManager.GetSelectedObject6(2, -1)
    MateData.EntitiesToMate(1) = SecondEntityToMate

    ' Set the Mate Alignment
    MateData.MateAlignment = 2
```

```
' Create the mate
Dim MateFeature As SldWorks.Feature
Set MateFeature = Part.CreateMate(MateData)
Part.ClearSelection2 True
Part.EditRebuild3
```

End Sub

Example

[Create a Cam-Follower Mate \(C#\)](#)

Remarks

A cam-follower mate:

- Allows you to mate a cylinder, plane, or point on a cam-follower part to the tangent faces of a cam part.
- Appears in the FeatureManager design tree as either CamMateCoincident or CamMateTangent.

[IMateFeatureData](#) is the parent of this mate interface.

To create a cam-follower mate:

1. Follow general instructions in the [IAssemblyDoc::CreateMate](#) Remarks.
2. Specify [ICamFollowerMateFeatureData::EntitiesToMate](#) for each EntityType or use [IModelDocExtension::SelectByID2](#) to pre-select the entities using Mark = 1 for the cam face and Mark = 8 for the cam-follower face or vertex.
3. Specify other properties of the CamFollowerMateFeatureData object.

To edit a cam-follower mate:

1. Access its feature and call [IFeature::GetDefinition](#) to get the MateFeatureData object.
2. Cast the MateFeatureData object to a CamFollowerMateFeatureData object.
3. Modify the CamFollowerMateFeatureData object.
4. Call [IFeature::ModifyDefinition](#).

To delete this cam-follower mate, use [IModelDocExtension::DeleteSelection2](#).

Access Diagram

[CamFollowerMateFeatureData](#)

See Also

[ICamFollowerMateFeatureData Members](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

ICamFollowerMateFeatureData Interface Members

[See Also](#) [Properties](#)



[SolidWorks.Interop.sldworks Namespace](#) : ICamFollowerMateFeatureData Interface

The following tables list the members exposed by [ICamFollowerMateFeatureData](#).

Public Properties

Name	Description
 EntitiesToMate	Gets or sets the entities to mate in this cam-follower mate.
 MateAlignment	Gets or sets the mate alignment of this cam-follower mate.

[Top](#)

See Also

[ICamFollowerMateFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

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ICamFollowerMateFeatureData Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : ICamFollowerMateFeatureData Interface

For a list of all members of this type, see [ICamFollowerMateFeatureData members](#).

Public Properties

Name	Description
------	-------------

 EntitiesToMate	Gets or sets the entities to mate in this cam-follower mate.
 MateAlignment	Gets or sets the mate alignment of this cam-follower mate.

[Top](#)

 See Also

[ICamFollowerMateFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

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EntitiesToMate Property (ICamFollowerMateFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICamFollowerMateFeatureData Interface](#) : EntitiesToMate Property (ICamFollowerMateFeatureData)

Gets or sets the entities to mate in this cam-follower mate.

 .NET Syntax

Visual Basic (Declaration)

```
Property EntitiesToMate( _  
    ByVal EntityType As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As ICamFollowerMateFeatureData  
Dim EntityType As System.Integer  
Dim value As System.Object  
  
instance.EntitiesToMate(EntityType) = value  
  
value = instance.EntitiesToMate(EntityType)
```

C#

```
System.object EntitiesToMate(  
    System.int EntityType  
) {get; set;}
```

C++/CLI

```
property System.Object^ EntitiesToMate {  
    System.Object^ get(System.int EntityType);  
    void set (System.int EntityType, System.Object^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

EntityType
Type of cam entity to mate as defined in [swCamMateEntityType_e](#) (see **Remarks**)

Property Value

Array of entities to mate (see **Remarks**)

 Visual Basic for Applications (VBA) Syntax

See [CamFollowerMateFeatureData.EntitiesToMate](#).

 Example

See the [ICamFollowerMateFeatureData](#) example.

 Remarks

Use this property with EntityType set to swCamMateEntityType_e.:

- CamPath, to get or set the cam [faces](#) to mate.
- CamFollower, to get or set the cam-follower face or [vertex](#) to mate.

Instead of specifying this property, you can use [IModelDocExtension::SelectByID2](#) to pre-select the entities to mate using Mark = 1 for the cam face and Mark = 8 for the cam follower face or vertex. You can pre-select mate entities during mate creation, but not during mate editing.

See Also

[ICamFollowerMateFeatureData Interface](#)
[ICamFollowerMateFeatureData Members](#)

Availability

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MateAlignment Property (ICamFollowerMateFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICamFollowerMateFeatureData Interface](#) : MateAlignment Property (ICamFollowerMateFeatureData)

Gets or sets the mate alignment of this cam-follower mate.

.NET Syntax

Visual Basic (Declaration)

Property MateAlignment As System.Integer

Visual Basic (Usage)



```
Dim instance As ICamFollowerMateFeatureData  
Dim value As System.Integer
```

```
instance.MateAlignment = value
```

```
value = instance.MateAlignment
```

C#

```
System.int MateAlignment {get; set;}
```

C++/CLI

```
property System.int MateAlignment {  
    System.int get();  
    void set ( & System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Mate alignment as defined in [swMateAlign_e](#)

Visual Basic for Applications (VBA) Syntax

See [CamFollowerMateFeatureData::MateAlignment](#).

Example

See the [ICamFollowerMateFeatureData](#) example.

See Also

[ICamFollowerMateFeatureData Interface](#)
[ICamFollowerMateFeatureData Members](#)

ICavityFeatureData (Type)

ICavityFeatureData Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : ICavityFeatureData Interface

Allows access to a cavity feature.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface ICavityFeatureData
```

Visual Basic (Usage)

```
Dim instance As ICavityFeatureData
```

C#

```
public interface ICavityFeatureData
```

C++/CLI

```
public interface class ICavityFeatureData
```

Visual Basic for Applications (VBA) Syntax

See [CavityFeatureData](#).

Example

[Create and Modify Cavity Feature \(C#\)](#)
[Create and Modify Cavity Feature \(VB.NET\)](#)
[Create and Modify Cavity Feature \(VBA\)](#)

Accessors

[IFeature::GetDefinition](#) and [IFeature::IGetDefinition](#)

Access Diagram

[CavityFeatureData](#)

See Also

[ICavityFeatureData Members](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ICoreFeatureData Interface](#)
[IFeatureManager::MoldUndercutDetect2 Method](#)
[IPartingLineFeatureData Interface](#)
[IPartingSurfaceFeatureData Interface](#)
[IRuledSurfaceFeatureData Interface](#)
[IShutOffSurfaceFeatureData Interface](#)
[IToolingSplitFeatureData Interface](#)
[IFeatureManager::InsertMoldCoreCavitySolids Method](#)
[IMoveFaceFeatureData Interface](#)

ICavityFeatureData Interface Members

[See Also](#) [Properties](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : ICavityFeatureData Interface

The following tables list the members exposed by [ICavityFeatureData](#).

Public Properties

Name	Description
 Components	Gets or sets the design components for this cavity.
 ScaleType	Gets or sets the point about which scaling occurs in this cavity feature.
 UniformScale	Gets or sets the value by which to scale this cavity feature in all directions.
 UseUniformScale	Gets or sets whether to uniformly scale this cavity feature.

[Top](#)

Public Methods

Name	Description
 AccessSelections	Gains access to the selections that define this cavity feature.
 GetComponentsCount	Gets the number of design components in this cavity feature.
 GetScale	Gets the values used to scale this cavity feature.
 IGetComponents	Gets the design components for this cavity feature.
 ISetComponents	Sets the design components for this cavity feature.
 ReleaseSelectionAccess	Releases access to the selections that define this cavity feature.
 SetScale	Sets the values by which to scale this cavity feature.

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See Also

[ICavityFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ICoreFeatureData Interface](#)
[IFeatureManager::MoldUndercutDetect2 Method](#)
[IPartingLineFeatureData Interface](#)
[IPartingSurfaceFeatureData Interface](#)
[IRuledSurfaceFeatureData Interface](#)
[IShutoffSurfaceFeatureData Interface](#)
[IToolingSplitFeatureData Interface](#)
[IFeatureManager::InsertMoldCoreCavitySolids Method](#)
[IMoveFaceFeatureData Interface](#)

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[Send comments on this topic.](#)

ICavityFeatureData Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : ICavityFeatureData Interface

For a list of all members of this type, see [ICavityFeatureData members](#).

Public Methods

Name	Description
 AccessSelections	Gains access to the selections that define this cavity feature.
 GetComponentsCount	Gets the number of design components in this cavity feature.
 GetScale	Gets the values used to scale this cavity feature.
 IGetComponents	Gets the design components for this cavity feature.
 ISetComponents	Sets the design components for this cavity feature.
 ReleaseSelectionAccess	Releases access to the selections that define this cavity feature.
 SetScale	Sets the values by which to scale this cavity feature.

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See Also

[ICavityFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ICoreFeatureData Interface](#)
[IFeatureManager::MoldUndercutDetect2 Method](#)
[IPartingLineFeatureData Interface](#)
[IPartingSurfaceFeatureData Interface](#)
[IRuledSurfaceFeatureData Interface](#)
[IShutoffSurfaceFeatureData Interface](#)

ICavityFeatureData Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : ICavityFeatureData Interface

For a list of all members of this type, see [ICavityFeatureData members](#).

Public Properties

Name	Description
Components	Gets or sets the design components for this cavity.
ScaleType	Gets or sets the point about which scaling occurs in this cavity feature.
UniformScale	Gets or sets the value by which to scale this cavity feature in all directions.
UseUniformScale	Gets or sets whether to uniformly scale this cavity feature.

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See Also

[ICavityFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ICoreFeatureData Interface](#)
[IFeatureManager::MoldUndercutDetect2 Method](#)
[IPartingLineFeatureData Interface](#)
[IPartingSurfaceFeatureData Interface](#)
[IRuledSurfaceFeatureData Interface](#)
[IShutOffSurfaceFeatureData Interface](#)
[IToolingSplitFeatureData Interface](#)
[IFeatureManager::InsertMoldCoreCavitySolids Method](#)
[IMoveFaceFeatureData Interface](#)

AccessSelections Method (ICavityFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICavityFeatureData Interface](#) : AccessSelections Method (ICavityFeatureData)

Gains access to the selections that define this cavity feature.

.NET Syntax

Visual Basic (Declaration)

```
Function AccessSelections( _  
    ByVal TopDoc As ModelDoc2, _  
    ByVal Component As Component2 _  
) As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As ICavityFeatureData  
Dim TopDoc As ModelDoc2  
Dim Component As Component2  
Dim value As System.Boolean  
  
value = instance.AccessSelections(TopDoc, Component)
```

C#

```
System.bool AccessSelections(  
    ModelDoc2 TopDoc,  
    Component2 Component  
)
```

C++/CLI

```
System.bool AccessSelections(  
& ModelDoc2^ TopDoc,
```



& Component2^ [Component](#)
)

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TopDoc

Top-level document (see **Remarks**)

Component

Component in which to modify the feature (see **Remarks**)

Return Value

True if the selections are successfully accessed, false if not

Visual Basic for Applications (VBA) Syntax

See [CavityFeatureData::AccessSelections](#).

Example

[Create and Modify Cavity Feature \(C#\)](#)

[Create and Modify Cavity Feature \(VB.NET\)](#)

[Create and Modify Cavity Feature \(VBA\)](#)

Remarks

To modify a feature in a...	Then...
Part	- TopDoc argument is the IModelDoc2 for the part - Component argument is NULL
Assembly	- TopDoc is the IModelDoc2 object for the assembly - Component argument is the IComponent2 object in which the feature is to be modified

This method puts the model into a rollback state to allow access to the selections that define this feature. You must use either of the following methods to restore the rollback state:

- [IFeature::ModifyDefinition](#) or [IFeature::IModifyDefinition2](#) if you modified the feature
- [ICavityFeatureData::ReleaseSelectionAccess](#) if you did not

See Also

[ICavityFeatureData Interface](#)

[ICavityFeatureData Members](#)

Availability

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Components Property (ICavityFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICavityFeatureData Interface](#) : Components Property (ICavityFeatureData)

Gets or sets the design components for this cavity.

.NET Syntax

Visual Basic (Declaration)

Property Components As System.Object

Visual Basic (Usage)



```
Dim instance As ICavityFeatureData  
Dim value As System.Object
```

```
instance.Components = value
```

```
value = instance.Components
```

C#

```
System.Object Components {get; set;}
```

C++/CLI

```
property System.Object^ Components {  
    System.Object^ get();  
    void set ( &    System.Object^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of design [components](#)

Visual Basic for Applications (VBA) Syntax

See [CavityFeatureData::Components](#).

Remarks

See [Accessing Selections that Define Features](#) for additional details on using this property.

See Also

[ICavityFeatureData Interface](#)
[ICavityFeatureData Members](#)
[ICavityFeatureData::GetComponentsCount Method](#)
[ICavityFeatureData::!GetComponents Method](#)
[ICavityFeatureData::!SetComponents Method](#)

Availability

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GetComponentsCount Method (ICavityFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICavityFeatureData Interface](#) : GetComponentsCount Method (ICavityFeatureData)

Gets the number of design components in this cavity feature.

.NET Syntax

Visual Basic (Declaration)

```
Function GetComponentsCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ICavityFeatureData  
Dim value As System.Integer
```

```
value = instance.GetComponentsCount()
```

C#

```
System.int GetComponentsCount()
```

C++/CLI

```
System.int GetComponentCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [CavityFeatureData::GetComponentCount](#).

Remarks

Call this method before calling [ICavityFeatureData::!GetComponentCount](#).

See Also

[ICavityFeatureData Interface](#)
[ICavityFeatureData Members](#)
[ICavityFeatureData::!SetComponents Method](#)
[ICavityFeatureData::Components Property](#)

Availability

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GetScale Method (ICavityFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICavityFeatureData Interface](#) : GetScale Method (ICavityFeatureData)

Gets the values used to scale this cavity feature.

.NET Syntax

Visual Basic (Declaration)

```
Sub GetScale( _  
    ByRef Xscale As System.Double, _  
    ByRef Yscale As System.Double, _  
    ByRef Zscale As System.Double _  
)
```

Visual Basic (Usage)



```
Dim instance As ICavityFeatureData  
Dim Xscale As System.Double  
Dim Yscale As System.Double  
Dim Zscale As System.Double  
  
instance.GetScale(Xscale, Yscale, Zscale)
```

C#

```
void GetScale(  
    out System.double Xscale,  
    out System.double Yscale,  
    out System.double Zscale  
)
```

C++/CLI

```
void GetScale(  
& [Out] System.double Xscale,  
& [Out] System.double Yscale,  
& [Out] System.double Zscale  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

XScale
Scaling value for the x direction
YScale
Scaling value for the y direction
ZScale
Scaling value for the z direction

Visual Basic for Applications (VBA) Syntax

See [CavityFeatureData::GetScale](#).

Example

[Create and Modify Cavity Feature \(C#\)](#)
[Create and Modify Cavity Feature \(VB.NET\)](#)
[Create and Modify Cavity Feature \(VBA\)](#)

See Also

[ICavityFeatureData Interface](#)
[ICavityFeatureData Members](#)
[ICavityFeatureData::SetScale Method](#)
[ICavityFeatureData::ScaleType Property](#)
[ICavityFeatureData::UniformScale Property](#)
[ICavityFeatureData::UseUniformScale Property](#)

Availability

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IGetComponents Method (ICavityFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICavityFeatureData Interface](#) : IGetComponents Method (ICavityFeatureData)

Gets the design components for this cavity feature.

.NET Syntax

Visual Basic (Declaration)

```
Sub IGetComponents( _  
    ByVal Count As System.Integer, _  
    ByRef Comps As Component2 _  
)
```

Visual Basic (Usage)



```
Dim instance As ICavityFeatureData  
Dim Count As System.Integer  
Dim Comps As Component2  
  
instance.IGetComponents(Count, Comps)
```

C#

```
void IGetComponents(  
    System.int Count,  
    out Component2 Comps  
)
```

C++/CLI

```
void IGetComponents(  
& System.int Count,  
& [Out] Component2^ Comps  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Count

Number of design components

Comps

- in-process, unmanaged C++: Pointer to an array of [IComponent2](#) objects of size count

- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Remarks

Call [ICavityFeatureData::GetComponentsCount](#) before calling this method.

See [Accessing Selections that Define Features](#) for additional details on using this method.

See Also

[ICavityFeatureData Interface](#)

[ICavityFeatureData Members](#)

[ICavityFeatureData::ISetComponents Method](#)

[ICavityFeatureData::Components Property](#)

Availability

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[Send comments on this topic.](#)

ISetComponents Method (ICavityFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICavityFeatureData Interface](#) : ISetComponents Method (ICavityFeatureData)

Sets the design components for this cavity feature.

.NET Syntax

Visual Basic (Declaration)

```
Sub ISetComponents( _  
    ByVal Count As System.Integer, _  
    ByRef Comps As Component2 _  
)
```

Visual Basic (Usage)



```
Dim instance As ICavityFeatureData  
Dim Count As System.Integer  
Dim Comps As Component2
```

```
instance.ISetComponents(Count, Comps)
```

C#

```
void ISetComponents(  
    System.int Count,  
    ref Component2 Comps  
)
```

C++/CLI

```
void ISetComponents(  
& System.int Count,  
& Component2^% Comps  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Count

Number of design components

Comps

- in-process, unmanaged C++: Pointer to an array of [IComponent2](#) objects of size Count

- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Visual Basic for Applications (VBA) Syntax

See [CavityFeatureData::ISetComponents](#).

Remarks

See [Accessing Selections that Define Features](#) for additional details on using this method.

See Also

[ICavityFeatureData Interface](#)
[ICavityFeatureData Members](#)
[ICavityFeatureData::IGetComponents Method](#)
[ICavityFeatureData::GetComponentsCount Method](#)
[ICavityFeatureData::Components Property](#)

Availability

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SOLIDWORKS API Help

[Send comments on this topic.](#)

ReleaseSelectionAccess Method (ICavityFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICavityFeatureData Interface](#) : ReleaseSelectionAccess Method (ICavityFeatureData)

Releases access to the selections that define this cavity feature.

.NET Syntax

Visual Basic (Declaration)

```
Sub ReleaseSelectionAccess()
```

Visual Basic (Usage)



```
Dim instance As ICavityFeatureData  
instance.ReleaseSelectionAccess()
```

C#

```
void ReleaseSelectionAccess()
```

C++/CLI

```
void ReleaseSelectionAccess();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [CavityFeatureData::ReleaseSelectionAccess](#).

Remarks

[ICavityFeatureData::AccessSelections](#) puts the model into a rollback state to allow access to the selections that define this feature.

Use this method to restore the rollback state if you did not modify the feature or use [IFeature::ModifyDefinition](#) or [IFeature::IModifyDefinition2](#) if you did.

See Also

[ICavityFeatureData Interface](#)
[ICavityFeatureData Members](#)

Availability

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SOLIDWORKS API Help

[Send comments on this topic.](#)

ScaleType Property (ICavityFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICavityFeatureData Interface](#) : ScaleType Property (ICavityFeatureData)

Gets or sets the point about which scaling occurs in this cavity feature.

.NET Syntax

Visual Basic (Declaration)

Property ScaleType As System.Integer

Visual Basic (Usage)



```
Dim instance As ICavityFeatureData
Dim value As System.Integer
```

```
instance.ScaleType = value
```

```
value = instance.ScaleType
```

C#

```
System.int ScaleType {get; set;}
```

C++/CLI

```
property System.int ScaleType {
    System.int get();
    void set ( & System.int value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Scale type as defined in [swCavityScaleType_e](#)

Visual Basic for Applications (VBA) Syntax

See [CavityFeatureData::ScaleType](#).

Example

[Create and Modify Cavity Feature \(C#\)](#)
[Create and Modify Cavity Feature \(VB.NET\)](#)
[Create and Modify Cavity Feature \(VBA\)](#)

See Also

[ICavityFeatureData Interface](#)
[ICavityFeatureData Members](#)
[ICavityFeatureData::GetScale Method](#)
[ICavityFeatureData::SetScale Method](#)
[ICavityFeatureData::UniformScale Property](#)
[ICavityFeatureData::UseUniformScale Property](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetScale Method (ICavityFeatureData)

[See Also](#)



Sets the values by which to scale this cavity feature.

.NET Syntax

Visual Basic (Declaration)

```
Sub SetScale( _
    ByVal Xscale As System.Double, _
    ByVal YScale As System.Double, _
    ByVal ZScale As System.Double _
)
```

Visual Basic (Usage)



```
Dim instance As ICavityFeatureData
Dim Xscale As System.Double
Dim YScale As System.Double
Dim ZScale As System.Double

instance.SetScale(Xscale, YScale, ZScale)
```

C#

```
void SetScale(
    System.double Xscale,
    System.double YScale,
    System.double ZScale
)
```

C++/CLI

```
void SetScale(
    & System.double Xscale,
    & System.double YScale,
    & System.double ZScale
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Xscale
Scaling value for the x direction
YScale
Scaling value for the y direction
ZScale
Scaling value for the z direction

Visual Basic for Applications (VBA) Syntax

See [CavityFeatureData::SetScale](#).

See Also

[ICavityFeatureData Interface](#)
[ICavityFeatureData Members](#)
[ICavityFeatureData::GetScale Method](#)
[ICavityFeatureData::ScaleType Property](#)
[ICavityFeatureData::UniformScale Property](#)
[ICavityFeatureData::UseUniformScale Property](#)

Availability

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Visual Basic (Declaration)

Property UniformScale As System.Double

Visual Basic (Usage)

```
Dim instance As ICavityFeatureData  
Dim value As System.Double
```

```
instance.UniformScale = value
```

```
value = instance.UniformScale
```

C#

```
System.double UniformScale {get; set;}
```

C++/CLI

```
property System.double UniformScale {  
    System.double get();  
    void set ( & System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Scaling value for all directions

Visual Basic for Applications (VBA) Syntax

See [CavityFeatureData::UniformScale](#).

Example

[Create and Modify Cavity Feature \(C#\)](#)
[Create and Modify Cavity Feature \(VB.NET\)](#)
[Create and Modify Cavity Feature \(VBA\)](#)

Remarks

Use [ICavityFeatureData::UseUniformScale](#) to enable uniform scaling of this cavity feature.

See Also

[ICavityFeatureData Interface](#)
[ICavityFeatureData Members](#)
[ICavityFeatureData::GetScale Method](#)
[ICavityFeatureData::SetScale Method](#)
[ICavityFeatureData::ScaleType Property](#)

Availability

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UseUniformScale Property (ICavityFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICavityFeatureData Interface](#) : UseUniformScale Property (ICavityFeatureData)

Gets or sets whether to uniformly scale this cavity feature.

Visual Basic (Declaration)

Property UseUniformScale As System.Boolean

Visual Basic (Usage)



```
Dim instance As ICavityFeatureData  
Dim value As System.Boolean
```

```
instance.UseUniformScale = value
```

```
value = instance.UseUniformScale
```

C#

```
System.bool UseUniformScale {get; set;}
```

C++/CLI

```
property System.bool UseUniformScale {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to use a uniform scaling value, false to not

Visual Basic for Applications (VBA) Syntax

See [CavityFeatureData::UseUniformScale](#).

Example

[Create and Modify Cavity Feature \(C#\)](#)
[Create and Modify Cavity Feature \(VB.NET\)](#)
[Create and Modify Cavity Feature \(VBA\)](#)

Remarks

Use [ICavityFeatureData::UniformScale](#) to set the uniform scaling value for this cavity feature.

See Also

[ICavityFeatureData Interface](#)
[ICavityFeatureData Members](#)
[ICavityFeatureData::GetScale Method](#)
[ICavityFeatureData::SetScale Method](#)
[ICavityFeatureData::ScaleType Property](#)

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